上海科技大学

2019-2020 学年第二学期本科生第一次期中考试考试卷

开课单位：物质学院

授课老师：李涛

考试：普通化学 II

课程代码：CHEM1111

姓名：_____

学号：_____

学院：_____

班级：_____

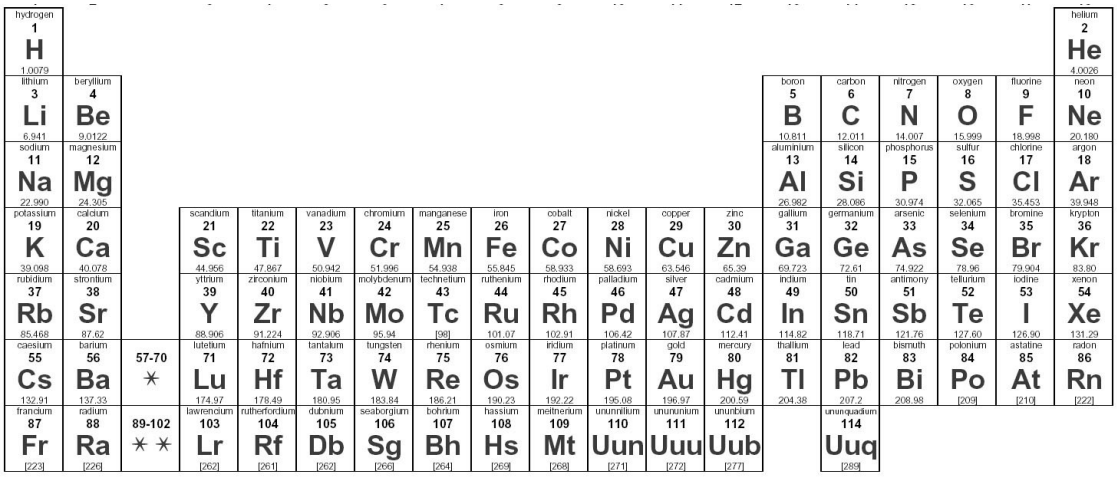
考生须知：

1. 请严格遵守考场纪律，禁止任何形式的作弊行为。
2. 答案请手写至答题纸上，并通过拍照或扫描上传。
3. 考试时间一小时，试卷提交时间5分钟。

考试成绩录入表：

题目	1	2	3	4	5	6	7	总分
记分								
复核								

评卷人姓名：　　日期：　　复核人姓名：　　日期：



1. (**15 pts, 3pts each**) Balance the following equations (add H_2O, OH^-, H_3O^+ if needed)

(a) $KClO_{3(aq)} + P_{(s)} \leftrightarrow KCl_{(aq)} + P_2O_{5(s)}$
(b) $C_2H_5OH_{(l)} + O_{2(g)} \leftrightarrow CO_{2(g)} + H_2O_{(l)}$
(c) $KNO_{3(aq)} + S_{(s)} + C_{(s)} \leftrightarrow K_2S_{(aq)} + N_{2(g)} + CO_{2(g)}$
(d) $P + Cu^{2+} \leftrightarrow Cu_3P + H_3PO_4$ (acidic environment)
(e) $UO_2^{2+} + Te_{(s)} \leftrightarrow U^{4+} + TeO_4^{2-}$ (acidic environment)

2. (**5 pts**) When 3.12 g of the nonvolatile solid anthracene, $C_{14}H_{10}$, is dissolved in 100.0 g of cyclohexane, C_6H_{12}, the boiling point of the cyclohexane is raised by 0.49°C. Calculate K_b for cyclohexane.

3. (**13 pts**) Imagine that two 1 L beakers, A and B, each containing an aqueous solution of fructose (a nonvolatile sugar with molecular weight = 180 g mol^{-1}) are placed together in a box, which is then sealed. (The concentrations of the solutions are not necessarily the same.) The temperature remains constant at 26°C. Initially, there is 600 mL of solution in A and 100 mL of solution in B. As the solutions stand in the sealed box, their volumes change slowly for a while. When they stop changing, beaker A contains 400 mL and beaker B contains 300 mL. It is then determined that the solution in A is 1.5 M in fructose and has a density of 1.10 g mL^{-1}.
(a) (**3 pts**) What is the molarity concentration of fructose in the solution in beaker B at the end? Explain.
(b) (**3 pts**) Calculate the concentration of fructose in the solution in A at the start.
(c) (**3 pts**) Calculate the concentration of the fructose in the solution in B at the start.
(d) (**4 pts**) The vapor pressure of pure water at 26°C is 25.2 torr. What is the pressure of water vapor in the box at the end, after the volumes have stopped changing?

4. (**15 pts**) Initially, 35.0 g oxygen is at a pressure of 2.00 atm and a temperature of 400 K. It expands adiabatically and reversibly until the pressure is reduced to 1.10 atm, and it is then compressed isothermally and reversibly until the volume returns to its original value. Calculate the final **pressure** and **temperature** of the oxygen, the **work** done and **heat** added to the oxygen in this process, and the **energy change** ΔU. Take $c_P(O_2) = 29.4$ J K^{-1} mol^{-1}.

5. (**20 pts**) A certain amount of SO_3 (g) is SEALED in a flask, and the temperature is raised to 550 K. The following equilibrium is established:

$$SO_3(g) \leftrightarrow SO_2(g) + 1/2\ O_2(g)$$

The total pressure in the flask when equilibrium is reached at this temperature is 0.835 atm, and the vapor density is 1.300 g L^{-1}.

(a) (**6 pts**) Calculate the partial pressure of each species.
(b) (**4 pts**) Calculate the equilibrium constant at this temperature.
(c) (**10 pts**) If additional O_2 was added to this equilibrated system which led to the rise of vapor density to 1.450 g L^{-1} at new equilibrium, what's the **final pressure** of the system?

6. (**20 pts**) In a titration experiment, a triprotic acid H_3AsO_4 solution (40.0 ml) was titrated with a 1.00 M NaOH solution. Using an appropriate indicator, the FIRST equivalence point was reached exactly after adding 50.0 ml of NaOH. ($K_{a1} = 5.00 \times 10^{-3}$; $K_{a2} = 9.30 \times 10^{-8}$; $K_{a3} = 3.00 \times 10^{-12}$)

(a) (**4 pts**) What is the concentration of the original H_3AsO_4 solution?

(b) (**6 pts**) What is the pH of the final solution?

(c) (**10 pts**) What are the concentrations of H_3AsO_4, $H_2AsO_4^-$, $HAsO_4^{2-}$, AsO_4^{3-} and OH^-?

7. (**12 pts**) Calculate the solubility of calcium oxalate (CuC_2O_4) in the following media.

(a) (**4 pts**) In pure water at 25°C.

(b) (**8 pts**) In 1.50 M oxalic acid ($H_2C_2O_4$) at 25°C ($K_{a1} = 5.9 \times 10^{-2}$, $K_{a2} = 6.4 \times 10^{-5}$ for $H_2C_2O_4$; $K_{sp} = 2.9 \times 10^{-8}$ for CuC_2O_4).